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Syllabus of New & Modified Courses for SUGC Presentation.docx

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Course Details for New BTech Mechanical Curriculum

New Courses:
1. The Art of Electronics
2. Differential Equations for Mechanical Engineers
3. Introduction to Engineering Materials
4. AI/ML for Mechanical Systems
5. Solid Mechanics Lab – 2
6. Fluid Mechanics Lab – 2
7. Design of Machine Elements: Lab
8. Systems Thinking
9. Controls Lab
10. Industrial Internet of Things (IIoT)
11. Automation Lab
12. Product Design & Realization
13. Special Topics in Mechanical and Aerospace Engineering

Updated / Modified Courses:
14. Fabrication Lab – 1
15. Fabrication Lab – 2
16. Introduction to Mechanical Systems
17. Thermodynamics
18. Solid Mechanics – 1
19. Solid Mechanics – 2
20. Solid Mechanics Lab – 1
21. Fluid Mechanics – 1
22. Fluid Mechanics – 2
23. Fluid Mechanics Lab – 1
24. Dynamics: Theory and Computations
25. Manufacturing Science - 1
26. Sensors and Signal analysis
27. Manufacturing Science - 2
28. Computer Aided Design and Manufacturing
29. Heat Transfer Lab
30. A Practical Introduction to FEM
31. A Practical Introduction to CFD
32. Control Systems
33. Industrial Engg. and Operations Research
34. Dynamics Lab

UG Electives
35. Introduction to Aerospace Vehicles
36. Flight Dynamics and Control
37. Industrial Automation
38. Circular Manufacturing
39. Digital Twins for Manufacturing
40. Lasers in Manufacturing
41. Physics and Principles of Micro- and Nano-Manufacturing
42. Power and Refrigeration Cycles

The Art of Electronics		
Fields	Values	Remarks
Date		
Course Code	ME1013	
Course Name	The Art of Electronics	
Credits	3	
Pre-Requisites (if any)	None	
Nature of Course	Combined theory and lab	
Type of Course	Core	
Targeted Program and Year	BTech ME, first year	
Contents	Kirchhoff's laws, Norton and Thevenin theorems, node method, SPICE simulations, AC analysis, impedances, frequency response, Bode plots, Filters, Analysis of nonlinear circuits, diodes DC and AC analysis, BJTs, MOSFETs,OPAMPS	
References	1. Horowitz and Hill, The art of electronics, Cambridge University Press, 2019. 2. Agarwal and Lang, Foundations of analog and digital electronic circuits, Elsevier, 2021. 3. Sedra and Smith, Microelectronic Circuits, Oxford University press, 2014 4. Behzad Razavi, Fundamentals of Microelectronics, Wiley, 2021. 5. Sherz and Monk, Practical electronics for inventors, McGraw Hill, 2016	
Course Objective	1. To make a ME student acquainted with basic electronics knowledge so they can communicate effectively with their electrical counterparts. 2. To give them enough background to design and debug simple circuits.	
Justification:	Currently, graduating mechanical engineers have no course which covers various topics on electronics. In real world is usually a combination of electronics and mechanical systems. Therefore a course which introduces them to the electronics is necessary.	

Differential Equations for Mechanical Engineers		
Fields	Values	Remarks
Date		
Course Code	ME1070	
Course Name	Differential Equations for Mechanical Engineers	
Credits	3	
Pre-Requisites (if any)	None	
Nature of Course	Theory	
Type of Course	Basic Sciences	
Targeted Program and Year	BTech-I	
Contents	• First Order ODE: Definition, Statement of existence and uniqueness solution, separable variables, homogeneous, non-homogeneous, linear, Bernoulli's equation, exact equation, Integrating factors • Second and Higher Order Equations: Homogeneous and non-homogeneous linear equations and linear systems with constant coefficients • Series Solutions: Power series method, Legendre Equation and Legendre polynomials, Frobenius method, Bessel's Equation and Bessel function, Sturm-Liouvelle problem, Orthogonal eigen function expansions • First Order PDE • Second Order PDE: Classification, Canonical forms • Wave (hyperbolic) Equation: DÁlemberts solution, Separation of variables, 2D wave equation • Heat (parabolic) Equation: 1D Unsteady heat flow, non-homogenous case • Laplace (elliptic) Equation: Steady heat flow in 2D, Polar coordinates, circular membrane, cylindrical and spherical coordinates	
	1. Boyce, Diprima and Meade, Elementary Differential Equations and Boundary Value Problems, 11th Edition, Wiley, 2017. 2. Jain & Iyengar, Advanced Engineering Mathematics, Fifth edition, Narosa Publications,	

References	2016 3. Erwin Kreyszig, Advanced Engineering Mathematics, 10th edition, J. Wiley and Sons, 2023 4. Peter V. O'Neil, Advanced Engineering Mathematics, Cengage, 2012. 5. Myint-U, Tyn, and Lokenath Debnath. Linear partial differential equations for scientists and engineers. Springer Science & Business Media, 2007. 6. 6. Strauss, Walter A. Partial differential equations: An introduction. John Wiley & Sons, 2007.	
Course Objective	This course aims to provide students with a background to solve various differential equations that are encountered in Mechanical and Aerospace Engineering.	
Justification:	Several courses for B.Tech in ME require an understanding of concepts in differential equations with examples relevant to ME education.	

Introduction to Engineering Materials

Fields	Values	
Date		
Course Code	ME1080	
Course Name	Introduction to Engineering Materials	
Credits	2	
Pre-Requisites (if any)	None	
Nature of Course	Theory	
Type of Course	Basic Engineering Skills (Interdisciplinary)	
Targeted Program and Year	BTech-I	
Contents	• Introduction to Materials and Crystallography: Interatomic bonding in materials; classification of engineering materials and their characteristic properties; unit cell and crystal systems; Crystallographic directions and planes (Miller indices); single crystal and polycrystalline metals; crystal structures: BCC, FCC, HCP. • Phase Diagrams, Phase Transformations, and Heat treatment: Concept of phases and Gibbs phase rule, binary phase diagrams: isomorphous, eutectic, and eutectoid systems; Iron–carbon equilibrium diagram and its significance in steels and cast irons; Kinetics of phase transformations: nucleation, grain growth, and interpretation of TTT and CCT diagram; Heat treatment: processes, related microstructural changes, and influence on mechanical properties • Plastic Deformation: Plastic deformation: dislocation, slip, and twinning; work hardening; recovery, recrystallization, and grain growth; cold working and hot working. • Mechanical behavior of Metals: Stress–strain behavior; Elastic and plastic deformation; mechanical properties: strength, ductility, hardness, toughness, and resilience; Fracture: ductile and brittle fracture • Engineering Ceramics, Polymers, and Composites • Structure, classification and applications	
References	1. Callister W.D., "Materials Science and Engineering" Wiley India, ISBN: 978-81-265-1076-4 2. Raghavan V. "Materials Science and Engineering- A first course." 5^{th} Edition, ISBN: 978-81-203-2455-8	
Course Objective	This course introduces the fundamental concepts of engineering materials from the perspective of appropriate material selection and understanding of the behaviour for Mechanical Engineering students. It familiarizes students with metals, alloys, ceramics, and polymers, with particular focus on their mechanical behavior. The course provides a foundation for material selection in engineering design and prepares students for advanced studies in manufacturing, solid mechanics, machine design and other related courses in mechanical engineering.	
Justification:	This course is essential for mechanical engineering students to understand the structure, properties, and behavior of materials, enabling informed material selection and design.	

AI/ML for Mechanical Systems

Fields	Values	Remarks
Date		
Course Code	ME2010	
Course Name	AI/ML for Mechanical Systems	
Credits	3	
Pre-Requisites (if any)	None	
Nature of Course	Theory	
Type of Course	Basic Engineering Skills	
Targeted Program and Year	BTech-II	
Contents	Brief history of AI, ML, and DL; machine learning models, interpretability - prediction vs inference; machine learning algorithms as optimization problems; supervised, unsupervised, semi-supervised learning approaches; regression and classification problems, encoding; linear and logistic regression; bias-variance tradeoff, overfitting and underfitting, regularization methods; feature scaling, feature selection, feature engineering; kNN, SVM, decision trees, ensemble methods; neural networks, backpropagation algorithm; application case studies in mechanical and aerospace engineering	
References	1. "An Introduction to Statistical Learning with applications in Python", James, Witten, Hastie, Tibshirani and Taylor; Springer, 2e, 2021. 2. "Probabilistic Machine Learning: An Introduction," Kevin P Murphy, 2022; MIT Press.	
Course Objective	To introduce students to the fundamentals of AI and ML concepts in an application-oriented way, enabling them to understand how machines can learn patterns from data relevant to mechanical systems. To demonstrate practical applications of AI/ML in mechanical engineering such as fault detection, condition monitoring, material property prediction, and design optimization, through hands-on exercises using sensor data and simple coding tools.	
Justification:	AI/ML techniques are being used in almost all areas of engineering. Most student projects nowadays involve usage of AI/ML techniques. This course introduces and lays the foundation of AI/ML with focus on application to mechanical and aerospace engineering problems.	

Solid Mechanics Lab – 2

Fields	Values	
Date		
Course Code	ME2141	
Course Name	Solid Mechanics Lab -2	
Credits	1	
Pre-Requisites (if any)	Solid Mechanics Lab -1	
Nature of Course	Lab	
Type of Course	Core	
Targeted Program and Year	B. Tech. II	
Contents	1. Euler Buckling 2. Thin Pressure Vessel 3. Uniaxial tension - Elastic and plastic behavior of Material 4. Failure of brittle and ductile materials 5. Wave propagation in bar	
References	1. S. Timoshenko (1956). Strength of Materials - II. D. Van Nostrand Company. 2. H. Kolsky (1953). Stress waves in solids. Dover publications.	
Course Objective	The aim of the course is to: 1. Test the applicability and limit of statements made and conclusions derived in the companion subject, Solid Mechanics - 2 2. Study the mechanical behavior of materials 3. Understand plasticity and failure 4. Analyze structural stability 5. Report writing.	
Justification:	The laboratory exercise is a 'hands-on' experience meant to show the relevance of theoretical concepts to understanding the behavior of real hardware and instrumentation and, at the same time, reveal how non-ideal conditions and some very 'un-theoretical' events can obscure the theoretical behavior.	

Fluid Mechanics Lab – 2

Fields	Values	Remarks
Date		
Course Code	ME2161	
Course Name	Fluid Mechanics Lab - 2	
Credits	1	
Pre-Requisites (if any)	Fluid Mechanics Lab – 1	
Nature of Course	Lab	
Type of Course	Core	
Targeted Program and Year	B.Tech II year	
Contents	Experiment List: 1 Flow measurement devices 2 Flow through pipes 3 Water channel experiment 4-6 DIY Experiment Suggestions for DIY Experiment: Students have to pick one experiment and execute it fully on their own. A few sample experiments are listed below. DIY - 1 DIY Flow meter Students can use any of the different types of flowmeter taught in the theory class or an entirely new one based on fluid mechanics principles and build a flow-meter on their own with the proposed idea. The flow meter needs to be calibrated and demonstrated against a known flow rate or independently validated using standard flow measurement set-ups in the laboratory. DIY - 2 Demonstrate the differences between Laminar and Turbulent Flows The experiment would involve building a channel flow set-up where the flow is visualised by injecting a dye in the channel. Depending on the flow characteristics (laminar/turbulent) the dye will exhibit a corresponding pattern. Students will have to build the channel set-up and demonstrate the two types of flows.	
References	1. Introduction to Fluid Mechanics, by Robert W. Fox and Alan T. McDonald, John Wiley & Sons 2. Fluid Mechanics, by Frank M. White, McGraw Hill 3. Fluid Mechanics by Kundu & Cohen 4. 4. Fluid Mechanics: Fundamentals and Applications by Yunus A. Cengel and John M. Cimbala, McGraw Hill	
Course Objective	1. To gain hands-on experience in applications of fluid mechanics principles. 2. To learn how to design and build an experimental setup using the concepts of fluid mechanics as an example.	
Justification:	Fluid Mechanics Lab in the present curriculum involves only 1 lab credit and it is difficult for students to get hands-on experience and som experience in building instrumentation. The proposed course is the second part of a two-part fluid mechanics lab course (each one credit), each of which provides basic hands-on experience in applying principles of fluid mechanics and designing set-ups.	

Design of Machine Elements: Lab

Fields	Values	Remarks
Date		
Course Code	ME3171	
Course Name	Design of Machine Elements: Lab	
Credits	1	
Pre-Requisites (if any)	Design of Machine elements (ME3170) course parallel being taken	
Nature of Course	Combined design and Lab	
Type of Course	Core	
Targeted Program and Year	B.Tech III	
Contents	1. Design of shaft subjected to combined constant twisting and bending loads 2. Design and selection of rolling element and journal bearing for a given rotating system specifications 3. Design of gear box for power transmission 4. Design of clutch and brake to control the power transmission 5. Design a transverse fillet welded joint subjected to a given tensile load. 6. To determine the fringe constant and stress concentration factor for the given Specimen using photo elastic test bench.	
References	1. 1. "Shigley"s Mechanical Engineering Design", Richards G. Budynas and J. Keith Nisbett, McGraw-Hill Education, 10th Edition, 2015. 2. Design of Machine Elements", V.B. Bhandari, Mc Graw Hill publishing company Ltd, 5th Edition 2020.	
Course Objective	• To practically relate to concepts discussed in Design of Machine Elements, • Students learn the translation of system specifications into component design	
Justification:	This lab course is new course and inline with learning by doing philosophy	

Systems Thinking

Fields	Values	Remarks
Date		
Course Code	ME3200	
Course Name	Systems Thinking	
Credits	1	
Pre-Requisites (if any)	None	
Nature of Course	Theory	
Type of Course	Basic Engineering Skills	
Targeted Program and Year	BTech-III	
Contents	Case-studies based course leading the journey from course learning to actual industry scenarios. With the help of case studies, the following aspects will be discussed: ● Interplay of various tools: experimental, simulation and fundamental physics of processes ● Interrelations of technology with users (human-centered thinking) ● New technology development and scaling (6Ds to Technology, S-Curve of Technology Development, Business Model Canvas) ● Techno-Commercial Readiness and Market Maturity Matrix: TCRM Matrix (Technology Readiness Level, Manufacturing Readiness Level, and Commercial Readiness Level)	
References	1. Hubert Anton Moser, "Systems Engineering, Systems Thinking, and Learning: A Case Study in Space Industry", Springer 2013. 2. NITI Aayog, "A New Lens for Innovation in New India - Introducing the Techno-Commercial Readiness and Market Maturity Matrix", NITI Aayog, 2023 3. Jamshid Gharajedhagi, System Thinking, Morgan Kaufhann, 2011.	
Course Objective	Set the tone for the journey of the student from course learning to actual industry scenarios.	
Justification:	As counterpart of Introduction to MAE course, this course intends to act as a culmination of varied learnings of the curriculum into a system level thinking.	

Controls Lab

Fields	Values	Remarks
Date		
Course Code	ME3211	
Course Name	Controls Lab	
Credits	1	
Pre-Requisites (if any)	Control Systems (course number ?)	
Nature of Course	Lab	
Type of Course	Core	
Targeted Program and Year	B. Tech. III	
Contents	Following exercises will be conducted as part of the laboratory course: ● Time response of first order and second order systems – Measuring time constants, observing step response, damping ratio, transient and steady state analysis ● First order system - Heater control ● Second order system - Spring Mass Damper System (Suspension System) ● Studying frequency response using Bode and Nyquist Plots ● Dynamic stability of a hanging bridge ● PID Controller Tuning using Ziegler-Nicholas Method ● PID control on a heating plate, record temperature response, and optimize controller gains using Ziegler-Nichols tuning rules The following exercises will be conducted over multiple lab sessions: ● System Identification and Transfer Function Estimation ● Estimation of acoustic transfer function of a duct ● Estimation of transfer function when a passive controller is incorporated ● Position control using a Servo Motor ● Robotic arm control ● 3D printer head control ● Control of Inverted Pendulum: Nonlinear system modelling, stability control ● Making a self-balancing robot ● Flight path control of a drone ● Sensing and Navigation	
References	1. Ogata, Katsuhiko. "Modern control engineering." (2020). 2. Nise, Norman S. Control systems engineering. John Wiley & Sons, 2020. 3. Franklin, Gene F., J. David Powell, and Abbas Emami-Naeini. Feedback control of dynamic systems. Vol. 10. Upper Saddle River, NJ: Pearson, 2010. 4. Thomson, William. Theory of vibration with applications. CrC Press, 2018.	
Course Objective	To introduce students to the application of concepts of control engineering related to mechanical systems Students will learn aspects of system identification, transfer function analysis, PID control tuning, and through their application in real world through both active and passive control design.	
Justification:	Currently, control system theory is taught in the department. Exposure to the experimental aspects of control systems design will enhance the learning experience of students.	

Industrial Internet of Things (IIoT)

Fields	Values	Remarks
Date		
Course Code	ME3250	
Course Name	Industrial Internet of Things (IIoT)	
Credits	1	
Pre-Requisites (if any)	None	
Nature of Course	Theory	
Type of Course	Core	
Targeted Program and Year	BTech-III	
Contents	● Introduction: Evolution of technology; progress from earlier industrial revolutions to Industry 4.0 ● Drivers of IIoT: Social & Economic Trends shaping I4, Mass Customization, Modularity, Ubiquitous digital systems. ● Enabling Systems: Sensors & IoT devices, Shared resources & cloud computing, Digital traceability and tamper proofing, Digital assets and security, Online process monitoring, Digital Twins, AR / VR ● Paradigms & Roadmap for future: Upgrading / retrofitting, VRIO Framework, Changing worker, Circular manufacturing	
	1. Ana Landeta Echeberria, "A Digital Framework for Industry 4.0", Palgrave Macmillan, 2020. 2. Alasdair Gilchrist, "Industry 4.0: The Industrial	

References	Internet of Things", 1st ed, Apress, 2017. 3. Sandra Grabowska and Sebastian Saniuk "Business Models for Industry 4.0: Concepts and Challenges in SME Organizations", Routledge, 2023 4. Jean-Claude André, "Industry 4.0 Paradoxes and Conflicts", Wiley, 2019.	
Course Objective	This course provides a comprehensive coverage on, among others, the role of machine-to-machine and human-to-machine communication, artificial intelligence, and digitalisation in manufacturing in the industry of tomorrow.	
Justification:	Give an orientation to students about the future and happening trends in the manufacturing sector, with the increased use of "smart" and "intelligent" machines.	

Automation Lab

Fields	Values	Remarks
Date		
Course Code	ME3351	
Course Name	Automation Lab	
Credits	1	
Pre-Requisites (if any)	None	
Nature of Course	Lab	
Type of Course	Core (BTech Mechanical only)	
Targeted Program and Year	BTech 3rd year	
Contents	The following is the list of experiments: 1. Elevator control using PLC 2. Conveyor belt control and object categorization using PLC 3. Pick and place tasks using robot manipulator 4. Assembly of parts using robot manipulator 5. Unipolar and bipolar stepper motor actuation using microcontroller board and raspberry pi 6. On/Off temperature control of a chamber using microcontroller board and raspberry pi.	
References		
Course Objective	The objective of this course is to (1) Introduce students to automation systems used in industry such as robot manipulators and Programmable Logic Controllers (PLCs) and (2) Train students to accomplish some of the tasks done by robot manipulators and PLCs using commercially off-the-shelf components.	
Justification:	Mechanical engineering is no longer limited to conventional design and manufacturing; automation and control are now integral parts of industry. Introducing automation concepts exposes students to the interdisciplinary nature of modern mechanical engineering, bridging mechanical systems, electronics, and computing.	

Product Design & Realization

Fields	Values	Remarks
Date		
Course Code	ME4063	
Course Name	Product Design & Realization	
Credits	3	
Pre-Requisites (if any)	None	
Nature of Course	Combined design and tutorial/lab	
Type of Course	Core	
Targeted Program and Year	BTech-IV	
Contents	● Introduction to Design thinking ● Brainstorming and Idea Selection ● System-Level Design ● Detail Design ● Component and subsystem manufacturing and Assembly ● Testing and Verification ● Prototype demonstration	
References	1. Karl T. Ulrich, Steven D. Eppinger, Maria C. Yang, " Product Design and Development", 7th Edition, 2020, McGraw Hill Publications 2. Edward B. Magrab, Satyandra K. Gupta et al., Integrated Product and Process Design and Development, 2nd ed, CRC Press 3. Karl T. Ulrich, Steven D. Eppinger, "Product Design and Development", 6th Edition, McGraw-Hill Education.	
Course Objective	To direct students toward the entire product design process process comprising of designing and development through visualization, planning and manufacturing of a product leading to 'Invention and Innovation'.	
Justification:	This course provides a unique hands-on project-based approach to integrate the learnings of designing, manufacturing and realization of mechanical engineering principles that the students would have learned in all their courses till before this course. The students will go through the entire design process through steps such as Visualize, Draw, Build, Improve, Modify, Identify, Suggest. Concept, Design (Mechanical, thermal, chemical), Drawing (2D/3D manufacturing details), Manufacturing, Testing, Simulation and Demonstration of final prototype.	

Fabrication Lab – 1

Fields	Values	Remarks
Date		
Course Code	ID1171	
Course Name	Fabrication Lab – 1	
Credits	2	
Pre-Requisites (if any)	None	
Nature of Course	Lab	
Type of Course	Basic Engineering Skills	
Targeted Program and Year	BTech-I	
Contents	1. Machine Shop – Introduction to lathe machine, tools and measuring instruments. Facing and Longitudinal turning, Step and taper turning, Chamfering and drilling. 2. Welding: Introduction to welding process, tools and machinery. Welding of any three joints - Single V-butt joint, Double lap Joint, Corner joint, T- joint, Edge joint, Gas cutting (Demo) 3. Electrical Shop - Wiring basics, Stair case wiring, Switch circuits, Flasher circuit, LDR 4. Electronics Shop: Soldering, wiring, resistor usage and symbols, Arduino programming, Laser based velocity sensor, battery charger	
References	1) Fundamentals Modern Manufacturing – Mikell P. Groover 2) Manufacturing Processes for Engineering Materials – Serope Kalpakjian, Steven Schmid	
Course Objective	This is a hands on training lab to be run in Central Workshop to equip students with the basic fabrication skills	
Justification:	This course provides hands-on experience working with various materials and tools, introducing students to fundamental fabrication techniques. This can help students design, prototype, and test their ideas.	

Fabrication Lab – 2

Fields	Values	Remarks
Date		
Course Code	ID1091	
Course Name	Fabrication Lab – 2	
Credits	2	
Pre-Requisites (if any)	None	
Nature of Course	Lab	
Type of Course	Basic Engineering Skills	
Targeted Program and Year	BTech-I	
Contents	1. Machine Shop - Introduction to general machines, Facing, Step turning, Drilling, Knurling, Boring, Taper turning, Thread Cutting (only Demo) 2. Welding - TIG Welding (Butt Joint with S.S.Plate), MIG welding (Butt Joint with M.S.Plate) 3. Laser engraving/cutting - CAD file formats, cutting and engraving, Laser-Cut 3D profiles 4. Polymer moulding - mould making with silicone, resin art, EPS foam cutting for pattern making	
References	1. Fundamentals Modern Manufacturing – Mikell P. Groover 2. Manufacturing Processes for Engineering Materials – Serope Kalpakjian, Steven Schmid	
Course Objective	This is a hands on training lab to be run in Central Workshop to equip students with the basic fabrication skills	
Justification:	This course provides hands-on experience working with various materials and tools, introducing students to fundamental fabrication techniques. This can help students design, prototype, and test their ideas.	

Introduction to Mechanical Systems

Fields	Values	Remarks
Date		
Course Code	ME1060	
Course Name	Introduction to Mechanical Systems	
Credits	3	
Pre-Requisites (if any)	None	
Nature of Course	Theory	
Type of Course	Basic Engineering Skills (Interdisciplinary)	
Targeted Program and Year	BTech First year	
Contents	An introduction to Mechanical Engineering; Disciplines of mechanical engineering; A quick tour of the broad areas of mechanical engineering: statics, dynamics, kinematics, thermodynamics, mechanical design, material science, computer-aided manufacturing and metallurgy, instrumentation; Research areas in mechanical engineering	
References	1. Manufacturing Processes for Engineering Materials – Serope Kalpakjian, Steven Schmid 2. S. Govindjee (2013). Engineering Mechanics of Deformable Solids. Oxford University Press. 3. Introduction to Fluid Mechanics, by Robert W. Fox and Alan T. McDonald, John Wiley & Sons 4. Introduction to sensors for electrical and mechanical engineers (1st edition, 2020)- Martin Novak, Taylor and Francis Publication. 5. Raghavan V. "Materials Science and Engineering- A first course." 5th Edition, ISBN: 978-81-203-2455-8	
Course Objective	This course is intended to give a broad introduction to students about Mechanical Engineering, its applications, its categories etc. for undergraduate students of all Engineering Branches.	
Justification:	Course need to provide a overview and quick glimpse of mechanical engineering.	

Thermodynamics

Fields	Values	Remarks
Date		
Course Code	ME2120	
Course Name	Thermodynamics	
Credits	3	
Pre-Requisites (if any)	None	
Nature of Course	Theory	
Type of Course	Core	
Targeted Program and Year	BTech II	
Contents	• Basic Concepts:- Systems and surroundings, Temperature and Zeroth Law, pressure, density. • Properties of Pure Substances:- Pure substance and its phases; Property evaluation, property diagrams and property tables; Ideal Gas equation of state; compressibility factor; Real Gas equations of state • Energy and 1st Law:- Energy and Energy transfer by Heat and Work; Various forms of work; 1st law of thermodynamics; Energy balance for closed systems; Internal energy, enthalpy, specific heats; Mass and energy balance for control volumes, flow work, steady state steady flow systems with examples. Energy efficiency. • Entropy and 2nd Law:- Heat Engines and the KP statement; Refrigerators, Heat Pumps and the Clausius statement; equivalency; reversible and irreversible processes; The Carnot cycle and Carnot heat engine, refrigerator and heat pump; Clausius Inequality and entropy as a system property; Increase of entropy principle; Entropy change for pure substances and ideal gases; the entropy change for processes; reversible steady flow work and isentropic efficiency; entropy balance equation for closed and open systems. • Basic thermodynamic cycles:- Carnot gas power cycle; Carnot vapor power cycle; the reversed Carnot cycle; Air standard assumptions for gas power cycles; Various gas power cycles:- Otto Cycle, Diesel cycle, Brayton cycle. Ideal Rankine vapor power cycle, Ideal Vapor-Compression Refrigeration Cycle; Heat Pump Systems; Gas Refrigeration Cycles.	
References	1. Thermodynamics: An Engineering Approach, 9th edition. Yunus Cengel, Michael Boles. 2. Fundamentals of Thermodynamics, 8th edition. Claus Borgnakke, Richard Sonntag	
Course Objective	Understand the basic application of 1st law and 2nd law of thermodynamics for design and performance evaluation of engineering devices, heat, power and refrigeration systems.	
Justification:	Only one core thermodynamics course is included instead of two as per earlier curriculum. Hence syllabus changed to include all the necessary fundamentals in a single course.	

Solid Mechanics – 1

Fields	Values	Remarks
Date		
Course Code	ME2130	
Course Name	Solid Mechanics -1	
Credits	2	
Pre-Requisites (if any)	None	
Nature of Course	Theory	
Type of Course	Core	
Targeted Program and Year	B. Tech. II	
Contents	• Fundamental Principles of Mechanics • Introduction to Mechanics of Deformable Bodies • Forces and Moments Transmitted by Slender Members • Stress and Strain • Mechanical Behaviour of Materials • Axial Load • Torsion • Bending • Deflection and Stress in Beams and Shafts • Buckling of Columns • Energy methods	
References	1. S. Govindjee (2013). Engineering Mechanics of Deformable Solids. Oxford University Press. 2. R. C. Hibbler (2018). Mechanics of Materials. Pearson. 3. S. H. Crandall, N. C. Dahl, and T. J. Lardner (2012). An Introduction to the Mechanics of Solids. Tata McGraw-Hill. 4. I. H. Shames and J. M. Pitarresi (2022). Introduction to Solid Mechanics. Prentice Hall.	
Course Objective	1. Understand the definition of internal forces, stress, and strain. 2. Develop a basic understanding of the mechanical properties of materials. 3. Stresses and deformation in axial load, torsion, and bending 4. Concepts for multiaxial states of stress 5. Gain proficiency in solving solid mechanics problems.	
Justification:	This is the first of the two core solid mechanics courses the MAE department offers undergraduate students. This course introduces the concepts of displacement, strain, stress, and constitutive equations of elastic solids in one, two, and three dimensions. Study material properties and solutions of one-dimensional structure members in tension, bending, and torsion.	

Solid Mechanics – 2

Fields	Values	Remarks
Date		
Course Code	ME2140	
Course Name	Solid Mechanics -2	
Credits	2	
Pre-Requisites (if any)	Solid Mechanics -1	
Nature of Course	Theory	
Type of Course	Core	
Targeted Program and Year	B. Tech. II	
Contents	1. Mathematical preliminaries 2. Theories of Deformation and Strain 3. Stress and Constitutive Relations 4. Boundary value problems in elasticity (Bending of Curved Beams, Torsion of Non-circular bars, plane stress/strain) 5. Energy Methods and Principles of Virtual Work 6. Dynamics: wave propagation and vibrations in elastic solids 7. Elastic Stability of structures 8. Theories of Failure 9. Introduction to Plasticity	
References	1. L S Srinath (2009). Advanced Mechanics of Solids. Tata McGraw-Hill. 2. I. H. Shames and J. M. Pitarresi (2022). Introduction to Solid Mechanics. Prentice Hall. 3. A. P. Boresi and R. J. Schmidt (2003). Advanced Mechanics of Materials. John Wiley & Sons.	
Course Objective	1. Understand the mathematical and physical foundations of the mechanics of solids, including deformation and stress measures, elastic and plastic stress-strain relations, and failure criteria. 2. Develop a basic understanding of materials' elastic and plastic properties. 3. Can formulate and solve boundary value problems in Solid Mechanics. 4. Ability to analyze wave propagation and vibrations in elastic solids. 5. Basic concepts for plasticity and Theories of Failure.	
Justification:	This is the second of the two core solid mechanics courses the MAE department offers undergraduate students. This course provides a firm foundation for the mechanics of deformable solids, enabling the student to analyze and solve various design problems encountered in practice.	

Solid Mechanics Lab – 1

Fields	Values	Remarks
Date		
Course Code	ME2131	
Course Name	Solid Mechanics Lab -1	
Credits	1	
Pre-Requisites (if any)		
Nature of Course	Lab	
Type of Course	Core	
Targeted Program and Year	B. Tech. II	
Contents	• Cable – Force-deflection relation • Uniaxial tension - Elastic Modulus of Material • Strain Gauge and Strain Measurement • Beam Bending • Torsion of Circular Shaft • Thermal Expansion of a bar	
References	1. D. Gross, W. Hauger, J Schroder, W A Wall and J Bonet (2018). Engineering Mechanics 2: Mechanics of Materials. Springer. 2. R. C. Hibbler (2018). Mechanics of Materials. Pearson.	
Course Objective	The aim of the course is to: 1. Test the applicability and limit of statements made and conclusions derived in the companion subject, Solid Mechanics - 1 2. Study force/displacement, behavior 3. Introduce measurement and instrumentation 4. Understand data analysis 5. Report writing.	
Justification:	The laboratory exercise is a 'hands-on' experience meant to show the relevance of theoretical concepts to understanding the behavior of real hardware and instrumentation and, at the same time, reveal how non-ideal conditions and some very 'un-theoretical' events can obscure the theoretical behavior.	

Fluid Mechanics – 1

Fields	Values	Remarks
Date		
Course Code	ME2150	
Course Name	Fluid Mechanics - 1	
Credits	2	
Pre-Requisites (if any)	None	
Nature of Course	Theory	
Type of Course	Core	
Targeted Program and Year	BTech II	
Contents	Differences between solids and fluids; introduction to continuum and continuum hypothesis; definition of fluid properties based on the concept of continuum- density, viscosity, surface tension; methods for describing fluid motion- Lagrangian and Eulerian descriptions; concepts of scalar and vector fields; material derivative; system and control volume Basic equations of fluid statics; manometer- U-tube manometer, inclined tube manometer, multi-liquid manometer; hydrostatic force on submerged surfaces- planar and curved surfaces; buoyancy; fluids in rigid body motion Reynolds Transport Theorem; integral form of conservation of mass for a control volume; integral form of conservation of linear momentum for an inertial control volume- stationary control volume and control volume moving with a constant velocity; integral form of conservation of linear momentum for an accelerating control volume Flow visualization- streamlines, streaklines and pathlines; Dimensional analysis	
References	1. Introduction to Fluid Mechanics, by Robert W. Fox and Alan T. McDonald, John Wiley & Sons 2. Fluid Mechanics, by Frank M. White, McGraw Hill 3. Fluid Mechanics by Kundu & Cohen 4. Fluid Mechanics: Fundamentals and Applications by Yunus A. Cengel and John M. Cimbala, McGraw Hill	
Course Objective	1. Basic concepts of fluid mechanics 2. Fundamentals of Fluid Statics; applications of fundamental concepts for solving problems 3. Integral (control volume) analysis of fluid flow problems; application to real-life problems 4. Introduction to flow visualization techniques and dimensional analysis	
Justification:	This will be the first of the two core fluid mechanics courses for undergraduate students that will be offered by the MAE department. We want to break up the entire fluid mechanics curriculum into two 2-credit courses so that the fundamental concepts can be covered in greater depth, with emphasis on developing problem-solving skills.	

Fluid Mechanics – 2

Fields	Values	Remarks
Date		
Course Code	ME2160	
Course Name	Fluid Mechanics - 2	
Credits	2	
Pre-Requisites (if any)	None	
Nature of Course	Theory	
Type of Course	Core	
Targeted Program and Year	BTech II	
Contents	Brief review and connection to integral analysis (FM I); Differential analysis- conservation of mass, continuity equation, concept of incompressible flows; conservation of linear momentum; Cauchy's equation of motion; Fluid kinematics- translation, rotation, angular deformation, linear deformation; constitutive behaviour of fluids; Navier-Stokes system of equations Solution of Navier-Stokes system of equations- plane Poiseuille flow; flow down an inclined plane; plane Couette flow; Hagen-Poiseuille flow (introduction to cylindrical co-ordinates); discussions on the different kinds of boundary conditions a brief discussion on Reynolds number; pressure scales; viscous flows and inviscid flows stream function; velocity potential; Euler's equations; Bernoulli equation and its application; potential flows Flow measuring devices- Pitot-static tube; venturimeter etc. Boundary layer theory- Description of the outer inviscid region; apparent contradiction from experiments; scaling analysis for the inner region; von Karman momentum integral equation; similarity analysis; Blasius' equation; lift and drag forces; concept of boundary layer separation	
References	1. Introduction to Fluid Mechanics, by Robert W. Fox and Alan T. McDonald, John Wiley & Sons 2. Fluid Mechanics, by Frank M. White, McGraw Hill 3. Fluid Mechanics by Kundu & Cohen 4. Fluid Mechanics: Fundamentals and Applications by Yunus A. Cengel and John M. Cimbala, McGraw Hill	
Course Objective	1. Differential analysis of fluid motion; introduction to Navier-Stokes system of equations 2. Solutions of Navier-Stokes system of equations for some classical fluid flow problems 3. Basics of scaling and non-dimensional analysis 4. Fundamentals of inviscid Flows 5. Overview of flow measurement devices 6. Fundamentals of boundary layer theory	
Justification:	This will be the second of the two core fluid mechanics courses for undergraduate students that will be offered by the MAE department. We want to break up the entire fluid mechanics curriculum into two 2-credit courses so that the fundamental concepts can be covered in greater depth, with emphasis on developing problem-solving skills.	

Fluid Mechanics Lab – 1

Fields	Values	Remarks
Date		
Course Code	ME2151	
Course Name	Fluid Mechanics Lab - 1	
Credits	1	
Pre-Requisites (if any)	None	
Nature of Course	Lab	
Type of Course	Core	
Targeted Program and Year	B.Tech II year	
Contents	1. Measurement of properties of fluids 2. Impact of a jet 3. Determination of centre of pressure, or, buoyancy and flotation 4-6. DIY Experiment Options for DIY Experiment: Students have to pick one experiment and execute it fully on their own. A few sample experiments are listed below. DIY - 1 Flow visualization past objects The students have to first propose what will be studied, make a presentation explaining the technical details and cost, provide a plan for execution, and execute the project. The grade will be based on their design, the experiments carried out by them, the presentation of the results DIY - 2 Cavitation in propellers as a demonstration of the effect of Bernoulli's equation	
	1. Introduction to Fluid Mechanics, by Robert W. Fox and Alan T. McDonald, John Wiley & Sons	

References	2. Fluid Mechanics, by Frank M. White, McGraw Hill 3. Fluid Mechanics by Kundu & Cohen 4. 4. Fluid Mechanics: Fundamentals and Applications by Yunus A. Cengel and John M. Cimbala, McGraw Hill	
Course Objective	1. To gain hands-on experience in applications of fluid mechanics principles. 2. To learn how to design and build an experimental setup using the concepts of fluid mechanics as an example.	
Justification:	Fluid Mechanics Lab in the present curriculum involves only 1 lab credit and it is difficult for students to get hands-on experience and som experience in building instrumentation. The proposed course is the first part of a two part fluid mechanics lab course (each 1 credit) each of which provides basic hands-on experience in applying principles of fluid mechanics as well as experience in designing set-ups.	

Dynamics: Theory and Computations

Fields	Values	Remarks
Date		
Course Code	ME2210	
Course Name	Dynamics: Theory and computation	
Credits	3	
Pre-Requisites (if any)	None	
Nature of Course	Theory	
Type of Course	Core	
Targeted Program and Year	B.Tech-I	
Contents	Review of numerical methods for differentiation and integrations; Introduction to Matlab, Solving equations of motion using built-in ODE solvers in Matlab; Particle dynamics in 2D and 3D, equations of motion for constrained particles in 2D and 3D using Newton's Law's, holonomic and non-holonomic constraints; Spring, damper and frictional elements; Energy conservation, Virtual work and Generalized forces, Equations of motions using Lagrange's principle; Case-studies: *Dzhanibekov effect* and spherical pendulum; Dynamics of Rigid body; 3D Rotations, Axis angle theorem, Euler parameterization of rotations, Angular velocity of a rigid body; Newton-Euler equations for a rigid body; Rigid body case-studies: rigid body suspended by a string, under gravity, rolling sphere, rolling coin, inter-connected rigid bodies	
References	1. Pratap, R. (2009). Getting started with MATLAB: a quick introduction for scientists and engineers. Oxford University Press, Inc. 2. Goldstein, Herbert. Classical mechanics. Pearson Education India, 2011.	
Course Objective	• To write equations of motion for mechanical systems comprising particles and 3D rigid bodies • To understand Newtonian and Lagrange approaches • To numerically solve the equation of motion, enabling simulation of the system	
Justification:	This course will introduce the concepts of dynamical systems. Students will learn the approaches to develop governing equations of motion for dynamical systems, and the numerical methods to solve these equations. Thereby, at the end of the course, the students are equipped with the fundamentals to analyse the dynamics of particles and rigid bodies.	

Manufacturing Science - 1

Fields	Values	Remarks
Date		
Course Code	ME2230	
Course Name	Manufacturing Science - 1	
Credits	3	
Pre-Requisites (if any)	None	
Nature of Course	Theory	
Type of Course	Core	
Targeted Program and Year	BTech-II	
Contents	Introduction to Manufacturing and its evolution, Net and near-net shape manufacturing; Metal Casting: Solidification of Alloys and its mechanism, Gating System Design and Estimation of Solidification time, Riser Design and Riser Placement, Defects, Other Solidification processes; Metal Forming: Mechanism of plastic deformation, fundamentals of plasticity, analysis of process, Defects, Rolling, Wire and tube drawing, Extrusion and Deep Drawing, other forming processes; Powder Processing (Metals and Ceramics); Polymer Part Manufacturing, Introduction and properties of polymer melts and Visco-elasticity, Processing of Thermoplastics (Extrusion, Injection Molding, Blow Molding, Rotational Molding) and Thermosets (compression and transfer molding); Additive Manufacturing, including applications for prototyping, tooling and manufacturing applications; Joining and Welding: Introduction, Solid State and Fusion Joining, Brazing and Soldering, Mechanical and Adhesive Property and Surface enhancement including heat treatment, surface treatment and coatings; Joining, Metal and nonmetal joining;	
References	1. Fundamentals Modern Manufacturing – Mikell P. Groover 2. Manufacturing Processes for Engineering Materials – Serope Kalpakjian, Steven Schmid 3. Manufacturing Science – Amitabha Ghosh; Asok Kumar Mallik	
Course Objective	This course along with Manufacturing Science - 2 aims to cover the basics of different manufacturing including its emphasis on the science of the processes. On successful completion of the course, the student is expected to have on the grasp on the following concepts 1. Appreciation for the importance of manufacturing techniques and a glimpse into the vast world of manufacturing techniques. 2. Solidification based manufacturing processes 3. Deformation based manufacturing processes 4. Joining techniques for different materials 5. Importance of additive manufacturing technologies	
Justification:	This is an entry level course which introduces the students to the different manufacturing methods. The emphasis is on the underlying fundamental physics of the processes and aims to foster a deeper appreciation into the multiphysics nature and the systems level approach of manufacturing technologies.	

Sensors and Signal analysis

Fields	Values	Remarks
Date		
Course Code	ME2253	
Course Name	Sensors and Signal Analysis	
Credits	3	
Pre-Requisites (if any)	None	
Nature of Course	Theory and Lab	
Type of Course	Core	
Targeted Program and Year	B. Tech. III	
Contents	Introduction to Measurement Fundamentals; Characteristics of Measurement Systems; Introduction to Sensors and Transducers Introduction to data acquisition, Microcontrollers, and serial communication; Sensing Principles for Key Measurements- Displacement, force, strain, velocity, acceleration, pressure, temperature, and motion; Sensor Performance Parameters- Range, Sensitivity, Accuracy, Repeatability, Noise; Fundamentals of Signal Processing; Time Waveform Analysis; Frequency Spectrum Analysis; Signal Filtering Techniques; Signal Conditioning Techniques; Advanced sensors: Optical sensors (LIDAR, laser displacement sensors), Smart sensors and IoT integration, Camera and RADAR fusion, IMU sensors	
References	1. Mechanical measurements (6th edition, 2018)-Thomas G Beckwith, Roy D Marangoni, John H Lienhard V, Pearson Publications. 2. Introduction to sensors for electrical and mechanical engineers (1st edition, 2020)- Martin Novak, Taylor and Francis Publication. 3. Digital signal processing (1st edition, 2015) - Alan V Oppenheim, Ronald H Schafer, Pearson Publications.	
Course Objective	This course provides an in-depth exploration of current industry practices while also offering insight into emerging trends in sensor technologies. It will deliver a thorough understanding of the fundamental principles behind sensor operation and demonstrate how these principles influence their application in real-world scenarios. Students will gain a solid foundation in signal analysis techniques and measurement methodologies, essential for engineering applications and data acquisition systems. The course will not only cover core concepts but also prepare students to adapt to evolving sensor technologies and their integration into modern systems.	
Justification:	Upon completing this course, students will be equipped to define and assess performance criteria for a variety of sensors, as well as predict and analyze the performance of different transducers and sensing technologies. They will gain a deep understanding of the underlying physics behind transduction mechanisms and develop the ability to interpret sensor specifications effectively. Additionally, students will be proficient in analyzing and interpreting sensor output data, and will be capable of designing sensor systems to address specific engineering challenges, applying their knowledge to propose optimal solutions.	

Manufacturing Science - 2

Fields	Values	Remarks
Date		
Course Code	ME3010	
Course Name	Manufacturing Science - 2	
Credits	3	
Pre-Requisites (if any)	No	
Nature of Course	Theory	
Type of Course	Core	
Targeted Program and Year	B.Tech III	
Contents	Conventional Removal and Finishing Processes: Importance of Material Removal and allied processes, classification; Chip Formation; Types of Chips; Tool Specification: Coordinate and Orthogonal Systems; Mechanics of Metal Cutting: Merchant's Circle Diagram, Stress, Strain and Strain Rate, determination of Shear Plane Angle; Tool Wear and Tool Life; Variables affecting Tool Life; Practical Machining Operations: Turning, drilling, milling; Finishing Operations: Grinding (MRR estimation, Wheel Specifications, Wheel Wear) and other processes; Unconventional Removal and Finishing Processes: Abrasive Jet Machining, Ultrasonic Machining; Electro Discharge Machining; Electro Chemical Machining; Abrasive Jet Machining; Electron Beam Machining; Laser Beam Machining, Finishing processes (AFM and other variants); Micro-Manufacturing and Nano-Manufacturing: Micro, Nano-Manufacturing Processes, Scaling laws with emphasis on micro-Manufacturing. Metrology: Tolerancing principles, Geometrical and dimensional measurements, Light based measurement techniques.	
References	1. Machining and Machine Tools by A.B. Chattopadhyay, 2nd edition, 2017 2. Metal Cutting Principles by Milton C. Shaw, 2nd edition, 2005 3. Nonconventional Machining by P. K. Mishra, 1997 4. Introduction To Machining Science, G.K.Lal, New Age International, 2007 5. Manufacturing Science – Amitabha Ghosh; Asok Kumar Mallik	
Course Objective	1. Mechanics and mechanism of conventional material removal processes 2. Basic concepts tool wear, tool life, and machining economics 3. Mechanics and mechanism of non-conventional material removal processes 4. Principles of micro-manufacturing and scaling laws	
Justification:	To enhance comprehensive learning and practical application, we propose increasing the course credits from two to three. This revision will allow students to engage more deeply with theoretical concepts and advanced machining techniques, ensuring a more thorough understanding of the subject matter and its academic significance.	

Computer Aided Design and Manufacturing

Fields	Values	Remarks
Date		
Course Code	ME3480	
Course Name	Computer Aided Design and Manufacturing	
Credits	3	
Pre-Requisites (if any)	No	
Nature of Course	Theory	
Type of Course	Core	
Targeted Program and Year	B.Tech III	
Contents	Introduction to computer graphics: Geometric modelling of curves (Hermite, Bezier, B-Spline,NURBS), Geometric modelling of surfaces (swept, parametric, freeform), Part modelling techniques: wireframe, surface and solid modelling, data representation and exchange formats, homogeneous coordinates and transformations. Reverse Engineering Techniques: contact and non-contact methods for measurements for 2D and 3D; point data and point cloud processing; interpolation and approximation methods. Computer Aided Manufacturing: fundamentals of part programming, path generation, post processing and verification; robot interpolation and kinematics.	

	Digital Twins: Computer aided process planning (CAPP), manufacturing process simulation, condition monitoring, virtual and distributed manufacturing.	
References	1. Mathematical elements for computer graphics by Rogers, David F., Adams, J. Alan 2. Geometric Modelling by Michael E. Mortenson 3. Procedural Elements for Computer Graphics by David F. Rogers 4. Reverse Engineering: An engineering perspective by Vinesh Raja, Kiran J. Fernandes	
Course Objective	1. On successful completion of the course, the student is expected to 2. Be equipped with basic knowledge in the computer based systems for control of manufacturing processes. 3. Learn geometric modelling techniques including curves, surfaces and solids 4. Learn the importance of process planning as a precursor to CAM 5. Be familiar with part programming 6. Learn reverse engineering methodologies using CAD/CAM as a tool.	
Justification:	CAD (Computer-Aided Design) enables engineers and designers to create precise and detailed models, while CAM (Computer-Aided Manufacturing) translates these designs into efficient production processes. Mastery of these tools is essential in modern manufacturing applications	

Heat Transfer Lab

Fields	Values	Remarks
Date		
Course Code	ME3111	
Course Name	Heat Transfer Lab	
Credits	1	
Pre-Requisites (if any)		
Nature of Course	Lab	
Type of Course	Core	
Targeted Program and Year	B. Tech. III	
Contents	1. Temperature measurement and calibration 2. Thermal conductivity of solids: Copper, Brass, and Aluminium 3. Determine the convective heat transfer coefficient of air in Free and Forced convection 4. Find the thermal conductivity of fluids 5. Determine the performance of a heat exchanger 6. Determine the emissivity and Stefan-Boltzmann constant 7. Determine the heat transfer coefficient in condensation	
References	1. Holman J. P., Heat Transfer, 10th Edition, 2017 2. Cengel, Y.A. and Ghajar, A.J., Heat and Mass Transfer, 5th Edition, 2017 3. Tropea, C., Yarin, A.L. and Foss, J.F., 2007. Springer handbook of experimental fluid mechanics	
Course Objective	The Heat Transfer Lab in the current curriculum includes only one lab credit, making it challenging to gain a proper understanding of heat transfer concepts and their applications in various thermal systems. Increasing it to two credits would allow sufficient time to develop deeper insights into the principles and their applications.	
Justification:	The Heat Transfer Lab in the current curriculum includes only one lab credit, making it challenging to gain a proper understanding of heat transfer concepts and their applications in various thermal systems. Increasing to two credits would allow sufficient time to develop deeper insights into the principles and their applications.	

A Practical Introduction to FEM

Fields	Values	Remarks
Date		
Course Code	ME3183	
Course Name	A Practical Introduction to FEM	
Credits	3	
Pre-Requisites (if any)	Solid-Mechanics I and II	
Nature of Course	Theory and Lab	
Type of Course	Core	
Targeted Program and Year	B. Tech. III	
Contents	• Overview of FEM • Differential Equations, Linear Algebra • Direct Stiffness Approach • Strong and Weak Form • Approximation of Trial Solution • Numerical Integration and Computer Implementation • 1-D Example Problems and Programming • Multidimensional Scalar Field • Isoparametric Elements • Multidimensional Vector Field • Dynamic Behaviour • Example Problems in Abaqus	
References	1. Jacob Fish and Ted Belytschko (2007). A first course in finite elements. John Wiley & Sons, Ltd. 2. T J R Hughes (2000). The Finite Element Method: Linear Static and Dynamic Finite Element Analysis. Dover Publications. 3. T. R. Chandrupatla and A. D. Belegundu (2022). Introduction to Finite Elements in Engineering. Cambridge University Press. 4. Singiresu S. Rao (2018). The Finite Element Method in Engineering. Elsevier.	
Course Objective	• The aim of the course is to introduce both the theory as well as practical aspects of finite element analysis. • Understand the fundamental theory of the finite element method (FEM). • Develop practical skills by implementing FEM in Python and using the commercial FEA software, ABAQUS. • Gain proficiency in solving solid mechanics problems. • Learn FEM problem formulation and solution strategies through hands-on lab assignments. • Apply FEM concepts to an engineering design challenge.	
Justification:	Partial differential equations are used as mathematical tools to study many physical phenomena in engineering and science. However, solving these equations analytically for arbitrary geometries is typically impractical. The finite element method (FEM) provides a numerical approach to obtain approximate solutions to these equations. In engineering, FEM is widely used for computer-based simulations of problems such as stress analysis, heat transfer, fluid flow, and electromagnetics.	

A Practical Introduction to CFD

Fields	Values	Remarks
Date	7th April 2025	
Course Code	ME3193	
Course Name	A Practical Introduction to CFD	
Credits	3	
Pre-Requisites (if any)	Fluid-Mechanics I and II	
Nature of Course	Theory and Lab	
Type of Course	Core	
Targeted Program and Year	B. Tech. 3rd year	
Contents	• Overview of CFD • Finite Difference approximation • Errors and order of accuracy • Finite difference approximation on non-uniform grids • Temporal discretization: Explicit and Implicit method • Linear Solvers: Direct (Gauss Elimination, TDMA) • Linear Solvers: Iterative (Gauss Jacobi, Gauss Siedel, Conjugate Gradient) • Types of PDEs and their characteristics • Solving Elliptic, Parabolic, and Hyperbolic PDEs numerically • Stability analysis of various discretization schemes • Solving Navier-Stokes Equation using Streamfunction-vorticity approach • Overview of MAC, SMAC, SIMPLE algorithm to solve Navier-Stokes Equation (Pressure-velocity coupling for incompressible flows) • Introduction to Finite volume method • Example problems in ANSYS Fluent/OpenFOAM	
References	1. John D. Anderson, JR. (1995). Computational fluid dynamics: the basics with applications. McGraw Hill Education New York 2. H. Versteeg and W. Malalasekra (2007). An introduction to computational fluid dynamics: The finite volume method. Pearson Education Ltd. 3. Dale A. Anderson, John C Tannehill, Richard H Pletcher, Ramakanth Munipalli, and Vijaya Shankar (2020). Computational Fluid Mechanics and Heat Transfer. CRC Press 4. Klaus A. Hoffmann and Steve T. Chiang (2000). Computational fluid dynamics volume I. Engineering Education system.	
Course Objective	• This course aims to introduce both the theoretical foundations and practical applications of Computational Fluid Dynamics (CFD). • Understand the fundamental concepts of Computational Fluid Dynamics (CFD), including governing equations and numerical discretization techniques. • Develop proficiency in numerical methods such as finite difference and finite volume for solving fluid dynamics problems. • Learn about error analysis, order of accuracy, and stability of numerical schemes to ensure reliable simulations. • Implement CFD algorithms in Python/Matlab/C/C++ • Apply direct and iterative linear solvers (e.g., Gauss Elimination, TDMA, Gauss-Seidel, Conjugate Gradient) to CFD problems. • Formulate and solve different types of PDEs (Elliptic, Parabolic, Hyperbolic) commonly encountered in fluid mechanics. • Understand pressure-velocity coupling algorithms such as MAC, SMAC, and SIMPLE for solving the Navier-Stokes equations. • Develop problem-solving skills in CFD applications through hands-on lab assignments.	
Justification:	Fluid dynamics is crucial in engineering and scientific applications, from aerodynamics and weather modeling to biomedical flows and industrial process simulations. The governing equations of fluid motion, such as the Navier-Stokes equations, are highly complex and often impossible to solve analytically for realistic geometries and boundary conditions. Computational Fluid Dynamics (CFD) provides a powerful numerical approach to approximate solutions for these equations. By leveraging discretization techniques like the finite difference and finite volume methods, one can simulate and analyze fluid behavior in various applications, including aerospace, energy, manufacturing, environmental engineering, etc.	

Control Systems

Fields	Values	Remarks
Date		
Course Code	ME3210	
Course Name	Control Systems	
Credits	2	
Pre-Requisites (if any)		
Nature of Course	Theory	
Type of Course	Core	
Targeted Program and Year	B. Tech. III	
Contents	Concept of control, stability and performance, modeling physical systems, Laplace transforms and transfer function, block diagrams, Routh's stability criterion, transient and steady state response specification and analysis, root locus analysis, lead, lag, and lead-lag compensator design; P, PI, PD, and PID controllers; modern control - state space representations of transfer function systems, controllability, observability, pole-placement; advanced concepts in control theory.	
References	1. K Ogata, "Modern Control Engineering," 5e, Pearson Education India, 2015. 2. Norman S Nise, "Nise's Control Systems Engineering," Wiley India, 4e, 2018. 3. Farid Golnarghi and Benjamin C. Kuo, "Automatic Control Systems," McGraw Hill Education (India), 10e, 2018.	
Course Objective	• Introduce concept of control and its applications in various domains like industrial automation, manufacturing, and robotics • Introduce the classical and modern approaches to control system design • Explain and emphasize the importance of (a) stability and (b) performance in control systems	
Justification:	The world is increasingly moving towards automation in all aspects of manufacturing and also in day-to-day life. This course will provide an essential foundation for advanced courses on industrial automation and robotics.	

Industrial Engg. and Operations Research

Fields	Values	Remarks
Date		
Course Code	ME3240	
Course Name	Industrial Engineering & Operations Research	
Credits	2	
Pre-Requisites (if any)	None	
Nature of Course	Theory	
Type of Course	Core	
Targeted Program and Year	BTech-III	
Contents	Basics of probability and statistics, Linear Programming Formulations, solutions, Transportation problems, Assignment problems, Networking problems, Dynamic programming, Queuing theory and its applications, Forecasting approaches, Monte Carlo simulation procedure (OR). Inventory models discussion (deterministic and probabilistic Models), Newsvendor model, Inventory Planning and Control, Economic Order Quantity (EOQ), Logistic, Distribution and Supply chain Management Work systems design: Concurrent engineering, Ranking system, Plant layout, Ergonomics, Cellular Manufacturing, Scheduling, Assembly Line Balancing. Quality management and control: Quality Improvement, Cost of Quality, Statistical Process Control, Central Tendency and Dispersion, Control Charts, Acceptance Sampling, New Quality Concepts, Preventive and breakdown maintenance approaches, reliability, Work study for Maintenance, Total Productive Maintenance (TPM), Spare Parts Management, Characteristics and classification of Spare parts; scheduling, layout design: Materials Requirement Planning (MRP), MRP-II, Enterprise Resource Planning (ERP).	
References	1. R. Dan Reid, Nada R. Sanders - Operations Management , Fourth Edition (2009, Wiley) 2. G Srinivasan, Operation research: Principles and Applications, 3rd Edition, PHI learning Private limited.	
Course Objective	● Build a foundation in optimization, probability, and statistics. ● Model and analyze industrial and service systems. ● Apply decision-making tools for efficient resource use. ● Use quantitative and computational methods for problem-solving. ● Develop a systems approach to improve productivity and sustainability.	
Justification:	The course on Industrial Engineering and Operations Research is essential for undergraduate students as it equips them with analytical and problem-solving skills required to tackle real-world industrial and service sector challenges. It provides a strong foundation in optimization, probability, and decision-making tools, enabling students to model, analyze, and improve complex systems. With growing emphasis on efficiency, sustainability, and data-driven decision-making, this course prepares students to design cost-effective and productive solutions. By integrating theory with practical applications, it develops competencies necessary for careers in manufacturing, supply chain, logistics, and management	

Dynamics Lab

Fields	Values	Remarks
Date		
Course Code	ME3371	
Course Name	Dynamics Lab	
Credits	1	
Pre-Requisites (if any)		
Nature of Course	Lab	
Type of Course	Core	
Targeted Program and Year	B. Tech. III	
Contents	Gear Efficiency Measurement, Planar Mechanism Demonstration, Rotary Balancing, Reciprocating Balancing, Static and Dynamic Analysis of Cam, Whirling of Shaft, Governors, Moment of Inertia Measurement, Vibrations, Rotational dynamics, Resonance, Kinematics experiments.	
References	1. Shigley et al. Theory of machines and mechanisms, Cambridge University Press, 2017 2. Goldstein, Herbert. Classical mechanics. Pearson Education India, 2011.	
Course Objective	The objective of the Dynamics Lab course is to enhance students' understanding of the principles of dynamics through hands-on experiments and practical applications. Students will analyze motion, forces, energy, and momentum in various mechanical systems, using experimental methods and data analysis techniques.	
Justification:	The Dynamics Lab in the current curriculum includes only one lab credit. The number of experiments are limited, thus making it challenging to gain a proper understanding of dynamics concepts and their applications.	

Special Topics in Mechanical and Aerospace Engineering

Fields	Values	Remarks
Date		
Course Code	ME4500	
Course Name	Special Topics in Mechanical and Aerospace Engineering	
Credits	2	
Pre-Requisites (if any)		
Nature of Course	Theory (Self Study) Pass/Fail	
Type of Course	Core	
Targeted Program and Year	B. Tech. IV	
Contents	Every year a basket of about 10 specialized topics will be floated by the faculty into which 4th year students can enroll. The course material (books, video lecture sources etc.) will be provided by the faculty. The courses will be of self-study nature. The faculty are in charge of evaluating the performance of the students in the course through evaluations/viva/assignments as appropriate.	
References	Topic dependent	
Course Objective	The objective of the course is to expose the students to emerging advanced and specialized topics that are part of the latest state-of-art in the field of Mechanical and Aerospace engineering but are not general enough to cover in a dedicated course.	
Justification:	The course will provide the students with the important skill set of self-motivated study and skill development. The students will gain the ability to gather and review disparate source materials and obtain a strong coherent foundation on a new subject area. Such abilities are essential in today's environment of continuous education and up-skilling.	

Introduction to Aerospace Vehicles

Fields	Values	Remarks
Date	Jul - Dec 2024	
Course Code	AE3010	
Course Name	Introduction to Aerospace Vehicles	
Credits	3	
Pre-Requisites (if any)	None	
Nature of Course	Theory	
Type of Course	Departmental Elective	
Targeted Program and Year	B Tech III IV	
Contents	• Introduction to Aerospace Engineering: Overview of aerospace vehicles - Historical development and milestones in aerospace engineering - Role of aerospace engineering in modern society • Principles of Flight: Basic aerodynamics principles - Forces acting on an aircraft • Lift, drag, thrust, and weight - Introduction to flight dynamics • Aerospace Propulsion Systems: Types of propulsion systems (jet engines, rockets, etc.) - Thermodynamics of propulsion - Propulsion performance parameters - Future trends in aerospace propulsion • Aircraft Design and Performance: Aircraft configuration and layout - Structural design considerations - Aircraft performance parameters (range, endurance, etc.) - Introduction to aircraft stability and control • Rotary-Wing Aircraft (Helicopters): Basics of rotary-wing aerodynamics - Helicopter design and operation - Helicopter performance parameters - Challenges and advancements in helicopter technology • Spacecraft Systems: Types of spacecraft (satellites, probes, manned spacecraft, etc.) - Orbital mechanics and trajectory analysis - Spacecraft subsystems (propulsion, communication, power, etc.) - Challenges and considerations in space mission design • Launch Vehicles and Rocket Propulsion: Basics of rocket propulsion - Launch vehicle design and operation - Launch vehicle performance parameters - Trends in space launch technology	
References	1. "Introduction to Flight" by John D. Anderson Jr., Ninth Edition, McGraw-Hill Education, 2020. 2. "Fundamentals of Aerospace Engineering" by Manuel Soler, First Edition, CreateSpace Independent Publishing Platform, 2014. 3. "Aerospace Engineering: From the Ground Up" by Ben Senson, First Edition, Delmar Cengage Learning, 2011. 4. "Rocket Propulsion Elements" by George P. Sutton and Oscar Biblarz, Ninth Edition, Wiley, 2016.	
Course Objective	This course provides an overview of various types of aerospace vehicles, including airplanes, helicopters, rockets, and spacecraft. Students will learn about the principles of flight, propulsion systems, structural design, and key technologies associated with different types of aerospace vehicles.	
Justification:	The philosophy of the "Introduction to Aerospace Vehicles" course emphasizes fostering a holistic understanding of aerospace technology. It aims to cultivate critical thinking by integrating theoretical concepts with practical applications, instilling an appreciation for the interdisciplinary nature of aerospace engineering. Through exploration of historical developments and contemporary challenges, students are encouraged to engage with ethical considerations and innovative solutions, preparing them for dynamic roles in the aerospace industry.	

Flight Dynamics and Control

Fields	Values	Remarks
Date	Jul - Dec 2024	
Course Code	AE3070	
Course Name	Flight Dynamics and Control	
Credits	3	
Pre-Requisites (if any)	None	
Nature of Course	Theory	
Type of Course	Elective	
Targeted Program and Year	BTech III, IV	
Contents	• Introduction to Flight Mechanics: Overview of aircraft motion in the atmosphere - Coordinate systems and reference frames - Aircraft components - Forces and moments on aircrafts - Equations of motion • Aerodynamics: Principles of aerodynamics - Lift and drag forces - Airfoil characteristics - Wing geometry and performance - Drag polar • Aircraft Performance (steady and accelerated flight): Climb, cruise, and descent performance - Thrust/Power required and Thrust/power available analysis - Range and endurance - V-n diagram - Takeoff and landing performance • Static Stability: Definition of stability and control - Longitudinal, lateral, and directional static stability - Neutral point and static margin • Dynamic Stability: Introduction to dynamic stability - Phugoid and short period modes - Dutch roll and spiral modes • Aircraft Control: Control surfaces and their functions - Control forces and moments - Control effectiveness and authority	
References	1. "Flight Stability and Automatic Control" by Robert C. Nelson, Second Edition, McGraw-Hill Education, 1998. 2. "Introduction to Flight" by John D. Anderson Jr., Ninth Edition, McGraw-Hill Education, 2020. 3. "Aircraft Performance and Design" by John D. Anderson Jr., First Edition, McGraw-Hill Education, 1999. 4. "Flight Dynamics Principles" by Michael V. Cook, Third Edition, Butterworth-Heinemann, 2013. 5. "Mechanics of Flight" by W. F Phillips, John Wiley and Sons, 2010	
Course Objective	This course introduces students to the fundamental principles of flight mechanics, covering topics such as aerodynamics, aircraft performance, stability, and control. Through theoretical study and practical examples, students will gain an understanding of the forces acting on an aircraft and how these forces influence its behavior in flight.	
Justification:	The Flight Mechanics course aims to provide students with a strong conceptual understanding of the principles governing aircraft behavior. It emphasizes the integration of theory and practice, critical thinking, and interdisciplinary approaches. Students learn to analyze complex flight scenarios, solve problems, and consider ethical and professional implications, preparing them for the dynamic challenges of the aviation industry.	

Industrial Automation

Fields	Values	Remarks
Date		
Course Code	ME4070	
Course Name	Industrial Automation	
Credits	3	
Pre-Requisites (if any)	None	
Nature of Course	Theory	
Type of Course	UG Elective	
Targeted Program and Year	BTech-III / IV	
Contents	• Introduction: Definition; Benefits of Automation; Types of automation: Fixed automation, programmable automation, and Flexible automation- Typical Features and examples; • Automated flow lines: types and their selection; Methods of workpart transport; Transfer mechanisms; Controlling of automated flow lines. • Automated assembly systems: Historical developments of the assembly process; Choice of assembly methods; Design for automated assembly; Transfer systems; Vibratory bowl feeders – its analysis; Non-vibratory feeders and their analysis; Analysis and design of part orienting devices, feed tracks and part placing mechanisms. • Sensors: study of various sensors required in a typical automated system for manufacturing. Construction and principle of operation of sensors. • Drives & Mechanisms: drives – types, selection criteria, construction and operating principle; Mechanisms: Ball screws, linear motion bearings, cams, systems controlled by camshafts; Transfer systems: Hydraulic, pneumatic & robotic. • Emerging automation paradigms and standards, including Industry 4.0 and Digital Twins.	
References	1. Groover, M. P., Automation, Production Systems, and Computer-Integrated Manufacturing, Prentice Hall, 2001. 2. Robots And Manufacturing Automation, 2Nd Ed, Ray Asfahl, Wiley India, 2010 3. G. Boothroyd – “Assembly Automation and Product Design”, Second Edition, Taylor & Francis, First Indian Edition - 2010	
Course Objective	To equip students with knowledge of automation principles, systems, and technologies, enabling them to design, analyze, and implement automated manufacturing systems.	
Justification:	This course prepares students towards the needs of Industry in manufacturing, automation, and robotics, by providing a comprehensive understanding of automation technologies, sensors, drives, and mechanisms.	

Circular Manufacturing

Fields	Values	Remarks
Date		
Course Code	ME4130	
Course Name	Circular Manufacturing	
Credits	1	
Pre-Requisites (if any)	None	
Nature of Course	Theory	
Type of Course	UG Elective	
Targeted Program and Year	BTech-III / IV	
Contents	• Linear vs. circular models in manufacturing. • The 6Rs of Manufacturing • Product Lifecycle & Design for Circularity • Resource Efficiency, lean, green, and circular synergies. • Circular Business Models & Policies	
References	1. Lerwen Liu · Seeram Ramakrishna “An Introduction to Circular Economy”, Springer, 2021 2. Peter Lacy, Jessica Long, Wesley Spindler “The Circular Economy Handbook”, Palgrave Macmillan, 2020	
Course Objective	To introduce the principles and practices of circular manufacturing, focusing on lifecycle thinking, resource efficiency, and sustainable business models.	
Justification:	This course equips learners with the knowledge and tools to transition from linear to circular manufacturing, enabling industries to reduce waste, optimize resources, and align with emerging sustainability policies.	

Digital Twins for Manufacturing

Fields	Values	Remarks
Date		
Course Code	ME4140	
Course Name	Digital Twins for Manufacturing	
Credits	1	
Pre-Requisites (if any)	None	
Nature of Course	Theory	
Type of Course	UG Elective	
Targeted Program and Year	BTech-III / IV	
Contents	• Introduction to Digital Twins: Evolution from simulation to digital twin; Definitions, scope, and significance in manufacturing; Types of digital twins (product, process, system, supply chain). • Enabling Technologies: IoT, sensors, real-time data acquisition and big data analytics. • Digital Twin in Manufacturing Operations: process monitoring and control; predictive maintenance, production scheduling and optimization • Challenges: interoperability, traceability, cybersecurity, data governance	
References	1. Digital Twin Driven Smart Manufacturing. 1st Edition, 2019. Imprint: Academic Press. Authors: Fei Tao, Meng Zhang, A.Y.C. Nee. 2. Digital Twins in Manufacturing by Vytautas Ostaševičius (2022)	
Course Objective	To understand the fundamentals of digital twin technology and its role in smart manufacturing.	
Justification:	This course equips learners with the knowledge and tools to leverage digital twins, inline with the evolving industry trends.	

Lasers in Manufacturing

Fields	Values	Remarks
Date		
Course Code	ME4150	
Course Name	Lasers in Manufacturing	
Credits	2	
Pre-Requisites (if any)	None	
Nature of Course	Theory	
Type of Course	UG and PG Elective	
Targeted Program and Year	BTech-III / IV / PG	
Contents	Laser design and basics of laser optics, Laser cutting, drilling and Piercing, Laser welding, Heat source modelling, laser surface treatment, Rapid Prototyping and Low-volume Manufacture, Laser Ablative Processes – Macro and Micromachining, Laser Bending or Forming, Laser cleaning, Monitoring and control in laser based manufacturing processes.	
References	1. William M. Steen, Jyotirmoy Mazumder, Laser Material Processing, 4th Edition, Springer. 2. W W Duley, Laser Processing and Anlaysis of Materials, 1st Edition, Springer.	
Course Objective	• Provide a strong foundation in laser physics, beam characteristics, and material–laser interactions. • Develop knowledge of laser-based processes such as cutting, welding, drilling, surface treatment, and additive manufacturing. • Enable students to analyze and optimize process parameters for precision, quality, and productivity. • Introduce modeling, simulation, and computational approaches for advanced laser processing applications. • Foster awareness of industrial applications, safety, and sustainability in laser-based manufacturing. • Encourage critical thinking and research skills to explore emerging trends and innovations in laser material processing. •	
Justification:	The course on Laser Material Processing is vital for both undergraduate and postgraduate students as it bridges fundamental science with advanced manufacturing applications. With lasers playing a critical role in cutting, welding, drilling, surface modification, and additive manufacturing, this course equips students with the knowledge to understand and optimize these processes. It fosters analytical and computational skills for modeling, simulation, and process control, ensuring precision and efficiency. By integrating safety, sustainability, and industrial relevance, the course prepares students for careers in aerospace, automotive, biomedical, and electronics industries, while also encouraging innovation and research in next-generation manufacturing technologies.	

Physics and Principles of Micro- and Nano-Manufacturing

Fields	Values	Remarks
Date		
Course Code	ME4160	
Course Name	Physics and Principles of Micro- and Nano-Manufacturing	
Credits	3	
Pre-Requisites (if any)	None	
Nature of Course	Theory	
Type of Course	UG Elective	
Targeted Program and Year	BTech-III / IV	
Contents	• Crystal Growth and Wafer Preparation • Lithography: Comparison between non-lithography and lithography-based manufacturing, Photolithography, Next-Generation Lithographies • Pattern transfer with Subtractive techniques: Dry Etching, Wet Chemical Etching and Wet Bulk Micromachining—Pools as Tools, Thermal Energy-Based Removing, Mechanical Energy-Based Removing • Pattern transfer with additive techniques: Physical and Chemical Vapor Deposition—Thin Film Properties and Surface Micromachining, Chemical, Photochemical, and Electrochemical Forming Techniques, Thermal Energy-Based Forming Techniques—Thermoforming, Micromolding Techniques—LIGA, Top down and bottom up manufacturing • Applications in the electronics industry (semiconductors, solar cells, sensors), biomedical field (stents, implants, drug-delivery devices), and other sectors such as aerospace, energy, photonics, and environmental technologies	
References	1. Fundamentals of Microfabrication and Nanotechnology, Volume II,Manufacturing Techniques for Microfabrication and Nanotechnology, Third Edition, Marc J. Madou, CRC Press 2. Fundamentals of Microfabrication and Nanotechnology : From MEMS to Bio-MEMS and Bio-NEMS: Manufacturing Techniques and Applications, 3rd Edition, Volume - III	
Course Objective	To provide students with a clear understanding of the physics and fundamental principles behind micro- and nano-manufacturing processes, including lithography, pattern transfer, and thin-film deposition, enabling them to analyze and apply these techniques for miniaturized systems.	
Justification:	Micro- and nano-manufacturing drives modern electronics, MEMS, and miniaturized devices. Understanding the underlying physics and principles is essential for designing, optimizing, and innovating advanced manufacturing processes.	

Power and Refrigeration Systems

Fields	**Values**	**Remarks**
Date		Course proposed date
Course Code	ME3160	In case of interdisciplinary course, it should be IDxxxx'.
Course Name	**Power and Refrigeration Systems**	
Credits	3	1, 2, 3, etc.
Pre-Requisites (if any)	Thermodynamics	Can be a Boolean Formula with ORs, ANDs and NOT (courses not to be taken along with this course)
Nature of Course	Theory	Theory, Lab, Design, Combined theory and lab, combined design and tutorial/lab, project /thesis, seminar
Type of Course	Departmental Elective	Core, Elective, interdisciplinary, etc.

Targeted Program and Year	BTech-III/IV	BTech (I, II, III or IV), B.Des, M.Sc, M.Tech, M.Des, MA, PhD, etc.
Contents	● Combined first and second laws. Availability and Exergy. Reversible work and irreversibility, Second law efficiency, Exergy Transfers and Exergy Balance Equations, Exergy Destruction and decrease of exergy principle, Exergy analysis of basic cycles. ● Gas Power Systems - Gaseous Working Fluids: Air-standard power cycles, Brayton Cycle, Brayton Cycle with Regeneration, Intercooling, Reheating, Gas turbine power cycles configurations, air standard cycle for Jet propulsion, Otto Cycle, Diesel Cycle, Stirling Cycle, Atkinson and Miller Cycles. Second law analysis of gas power cycles ● Vapor Power Systems - with Phase Change: Rankine cycle, Effect of pressure temperature on Rankine cycle, Reheat cycles, regenerative cycle, Second Law analysis of Vapor Power Cycles, Cogeneration, Combined Cycles ● Refrigeration Systems:- Vapour Compression Refrigeration Cycle, working fluids for vapour compression refrigeration systems, deviation of actual cycles from ideal cycle, second law analysis of vapor-compression cycles, heat pump systems, cascade and multi-stage refrigeration systems, gas refrigeration systems and cycles, the absorption refrigeration cycle, combined cycle refrigeration systems. ● Gas Mixtures: General considerations of mixture of ideal gases, Mixture model involving gases and vapor, Energy equation applied to gas vapor mixtures, adiabatic saturation process, Wet bulb and dry bulb temperatures and Psychometric chart. ● Thermodynamic Relations: Clapeyron equation, mathematical relations for homogeneous phase, Maxwell relations, thermodynamic relations involving enthalpy, internal energy, and entropy, property relations for mixtures, pseudopure substance models for real gas mixtures. ● Reactive Systems Thermodynamics:- Fuels and Combustion, Enthalpy of Formation, 1^{st} law analysis of reacting systems: Enthalpy of Combustion, Heat of Reaction and Adiabatic Flame Temperature; Second Law Analysis of Reacting Systems; Fuel Cells; Criteria for Chemical Equilibrium; Equilibrium Constant for Ideal Gas Mixtures; Equilibrium analysis for simultaneous reactions; Phase Equilibrium: Single and multi-component systems	
References	1. Thermodynamics: An Engineering Approach, 9th edition. Yunus Cengel, Michael Boles. 2. Fundamentals of Thermodynamics, 8th edition. Claus Borgnakke, Richard Sonntag	Minimum two references have to be provided along with the Year and Edition
Course Objective	The course aims to provide Mechanical Engineering students a comprehensive understanding of the various thermodynamic cycles used in power and refrigeration systems and how to analyze their performance using 1^{st} and 2^{nd} law. It also exposes students to the thermodynamics of reactive systems that are used in combustion and fuel cell applications. After taking the course, the students are expected to become well versed in analysis of various types of power and refrigeration cycles used in energy production, HVAC industry and transportation sectors.	List of items student should know on passing the course
Justification:	Syllabus edited to reflect that it is now a B Tech elective (and not a core) course in the UG curriculum.	1) Philosophy of the course 2) If the course matches in content with another course, please provide comparison. Any two courses should not match largely